

Method (2)

Q ① solve $x(y-z)p + y(z-x)q = z(x-y)r$

Sol here $P = x(y-z)$, $Q = y(z-x)$,
 $R = z(x-y)$

Lagrange's auxiliary eqn is

$$\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$$

using $l=1, m=1, n=1$ as multipliers

$$\Rightarrow \text{then } \frac{1 \cdot dx + 1 \cdot dy + 1 \cdot dz}{\cancel{xy} - \cancel{xz} + \cancel{yz} - \cancel{xy} + \cancel{xz} - \cancel{yz}} = 0$$

$$\text{so } dx + dy + dz = 0$$

$$\Rightarrow x + y + z = c_1 \quad \text{--- (1)}$$

$$\text{Now take } l = \frac{1}{x}, m = \frac{1}{y}, n = \frac{1}{z}$$

$$\text{i.e. } \frac{1}{x} \cdot \cancel{x(y-z)} + \frac{1}{y} \cdot \cancel{y(z-x)} + \frac{1}{z} \cdot \cancel{z(x-y)} = 0$$
$$y - z + z - x + x - y = 0$$

$$\text{so } \frac{1}{x} \cdot dx + \frac{1}{y} \cdot dy + \frac{1}{z} \cdot dz = 0$$

$$\Rightarrow \log x + \log y + \log z = \log c_2$$
$$xyz = c_2$$

$$\text{thus } \phi(x + y + z, xyz) = 0 \text{ Ans.}$$

Q 8 Solve $(x^2 - y^2 - z^2)p + 2xyzq = 2xz$

Sol Lagrange's auxiliary eqn is

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$$

$$\Rightarrow \frac{dy}{2xy} = \frac{dz}{2xz}$$

$$\rightarrow \log y = \log z + \log c$$

$$\Rightarrow y = cz \rightarrow c = \frac{y}{z} \quad \text{--- (1)}$$

Now taking multipliers x, y & z

$$\frac{x dx + y dy + z dz}{x^3 - xy^2 - xz^2 + 2xy^2 + 2xz^2} = \frac{dx}{2xz}$$

$$\Rightarrow \frac{x dx + y dy + z dz}{x(x^2 + y^2 + z^2)} = \frac{dx}{2xz}$$

$$\Rightarrow \frac{2(x dx + y dy + z dz)}{2x(x^2 + y^2 + z^2)} = \frac{dx}{2xz}$$

on integrating

$$\int \frac{2(x dx + y dy + z dz)}{x^2 + y^2 + z^2} = \int \frac{dx}{xz}$$

$$x + y + z, xyz = 0 \text{ Ans.}$$

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$$\log(x^2 + y^2 + z^2) = \log z + \log C_2$$

$$\frac{x^2 + y^2 + z^2}{z} = C_2$$

hence the complete soln is

$$f\left(\frac{x^2 + y^2 + z^2}{z}, \frac{y}{z}\right) = 0.$$

③ Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

4) Solve
 $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2).$